

Multi-View Body Condition Scoring in Dairy Cattle

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Abstract. Body Condition Score (BCS) is a key indicator for assessing the health, productivity, and nutritional status of dairy cattle. Traditional methods rely on manual evaluation, which is subjective, time-consuming, and difficult to scale. In this work, a multi-view deep learning framework is proposed for automated BCS estimation using image data. The system utilizes two viewpoints top-view and back-view to capture anatomical features of cattle. A classification-based EfficientNet-B0 model is employed for both top-view and back-view images. Transfer learning with fine-tuning of the final classification layer is applied to adapt the pretrained models for Body Condition Score (BCS) estimation. The models are trained on publicly available datasets and evaluated using real-world farm data, with ground truth BCS provided by a veterinarian. The predictions from both viewpoints are compared to analyze the effectiveness of viewpoint-specific learning for BCS estimation. Experimental results achieved approximately 69 percent validation accuracy for the top-view model and 78 percent validation accuracy for the back-view model, indicating that back-view images provide more discriminative anatomical features for BCS prediction.

Keywords: Body Condition Score · Deep Learning · EfficientNet · Multi-View Learning · Precision Livestock Farming · Agri-CPS

1 Introduction

Body Condition Score (BCS) is a widely used indicator for assessing the health, productivity, and nutritional status of dairy cattle. It reflects the level of fat reserves in an animal and is closely associated with milk yield, reproductive performance, and overall herd management. Maintaining an optimal BCS is essential for ensuring animal welfare and improving farm efficiency in modern dairy systems [2, 3].

Traditionally, BCS is evaluated manually by veterinarians or trained personnel through visual inspection and palpation. Although widely adopted, this approach suffers from several limitations, including subjectivity, inter-observer variability, and lack of scalability in large-scale farming environments. Frequent manual evaluation is time-consuming and impractical for continuous monitoring, which limits its effectiveness in modern dairy operations [8].

With the advancement of precision livestock farming and Agricultural Cyber-Physical Systems (Agri-CPS), there is increasing emphasis on integrating physical farm environments with intelligent data-driven systems. In this context, computer vision and deep learning techniques have emerged as promising solutions for automated animal assessment. Convolutional Neural Networks (CNNs) have demonstrated strong capability in extracting complex visual features from images, making them suitable for BCS estimation tasks. Several studies have applied deep learning models for BCS prediction, achieving improved performance compared to traditional approaches [1, 2, 4].

Despite these advancements, most existing approaches rely on a single viewpoint of the animal, which may fail to capture all relevant anatomical features required for accurate scoring. Different viewpoints provide complementary information about body structure. For example, the top view captures overall body shape and spine curvature, while the back view highlights features such as the tail head and hip bones. Leveraging multiple viewpoints can therefore improve prediction robustness and accuracy [5, 6]. To ensure architectural consistency and computational efficiency, EfficientNet-B0 was selected for both viewpoints and fine-tuned using transfer learning.

Motivated by these limitations, this work proposes a multi-view deep learning framework for automated BCS estimation in dairy cattle. The system employs two independent models trained on top-view and back-view images, respectively. The predictions are evaluated using real-world farm data and compared with veterinarian-provided ground truth scores, enabling a systematic analysis of viewpoint effectiveness and model performance in practical agricultural environments.

The primary contribution of this work is a viewpoint-wise comparative analysis using a unified EfficientNet-B0 architecture under real-world farm conditions.

2 Related Work

Body Condition Score (BCS) estimation has been widely studied in the context of precision livestock farming, as it serves as an important indicator of animal health and productivity. Traditional approaches relied on manual assessment, which is subjective and labor-intensive. To overcome these limitations, researchers have explored automated methods using computer vision and machine learning techniques.

Early approaches primarily utilized classical image processing methods and handcrafted features to estimate BCS. However, these methods were limited in their ability to capture complex visual patterns and were highly sensitive to variations in lighting, background, and pose, resulting in poor generalization in real-world conditions.

With the advancement of deep learning, Convolutional Neural Networks (CNNs) have become the dominant approach for image-based BCS estimation. Rodríguez Alvarez et al. demonstrated the use of CNNs with depth images to capture structural features of cattle more effectively [1]. Similarly, Liu et al. pro-

posed an ensemble-based approach that combines multiple models to improve prediction accuracy [2]. Recent work by Nagy et al. and Çevik further highlighted the effectiveness of deep learning models in achieving fine-grained BCS classification and real-time deployment in farm environments [3, 4].

Beyond BCS estimation, deep learning has been extensively applied to various livestock monitoring tasks. For instance, CNN-based methods have been used for cattle identification using facial and body features, enabling automated tracking of individual animals. Activity recognition systems have been developed to monitor behaviors such as feeding, walking, and resting, which are important indicators of animal health and welfare. Furthermore, deep learning approaches have been applied to disease detection, including lameness identification and early detection of mastitis, using visual and sensor-based data. These studies demonstrate the broader applicability of deep learning in precision livestock farming and highlight its effectiveness in extracting meaningful features from complex biological patterns.

Most existing BCS estimation approaches rely on a single viewpoint of the animal, typically using back-view images. However, a single viewpoint may not capture all relevant anatomical features required for accurate scoring. To address this limitation, multi-view approaches have been explored. Wu et al. demonstrated that combining multiple viewpoints can improve performance in cattle analysis tasks [5]. Similarly, Xu et al. showed that multi-view learning enhances robustness and generalization in livestock monitoring systems [6].

Despite these advancements, several challenges remain. Many existing systems are trained and evaluated on controlled datasets and lack validation in real-world farm conditions. In addition, there is limited work comparing different viewpoints to determine their relative effectiveness for BCS estimation.

This work addresses these limitations by proposing a multi-view deep learning framework and validating it using real-world farm data with expert-provided ground truth, enabling a more practical and comprehensive evaluation.

Unlike prior approaches that utilize different architectures for different viewpoints, this work employs a unified EfficientNet-B0 architecture for both views to enable fair comparison between viewpoints.

3 Problem Statement

The assessment of Body Condition Score (BCS) in dairy cattle is essential for monitoring animal health and improving farm productivity. However, traditional manual evaluation methods are subjective, time-consuming, and difficult to scale for large herds. Although recent deep learning-based approaches have shown promising results, most existing systems rely on a single viewpoint and are often evaluated under controlled conditions, limiting their applicability in real-world environments [5, 6].

In this work, the problem is defined as the development of an automated system for estimating BCS using image data, while addressing the limitations of existing approaches in terms of viewpoint dependency and real-world validation.

Input:

- A top-view image of the animal
- A back-view image of the same animal

Output:

- Predicted BCS value from the top-view model
- Predicted BCS value from the back-view model

These predictions are evaluated against the ground truth BCS assigned by a veterinarian, enabling a comparative analysis of model performance and the effectiveness of different viewpoints in real-world conditions.

Furthermore, the system analyzes prediction outcomes under different scenarios, including complete agreement, partial agreement, and disagreement with expert-provided scores. This enables a comprehensive evaluation of model reliability and practical applicability in agricultural settings.

4 Methodology

4.1 System Overview

The proposed system estimates the Body Condition Score (BCS) of dairy cattle using a multi-view deep learning framework. For each animal, two images are considered: a top-view image and a back-view image. Each image is processed independently using a dedicated deep learning model trained for that specific viewpoint.

The system produces two outputs: – BCS_t from the top-view model – BCS_b from the back-view model

These predictions are compared with the veterinarian-provided ground truth to evaluate model performance under real-world conditions.

4.2 Model Architecture

Separate EfficientNet-B0 models are trained independently for each viewpoint:

Top-View Model: Trained using top-view cattle images to learn features such as spine visibility, body width, and overall fat distribution.

Back-View Model: Trained using back-view cattle images to capture features around the tail head, hip bones, and rear body structure.

Both models are trained using supervised learning, where each image is associated with a ground truth BCS label. Both models utilize transfer learning with ImageNet-pretrained weights. Initially, pretrained convolutional layers are used as feature extractors, while the final classification layer is replaced according to the number of BCS classes.

Fine-tuning is performed on the final classification layer while retaining pre-trained feature representations from EfficientNet-B0. This approach reduces training complexity and improves convergence on limited agricultural datasets.

Using the same architecture for both viewpoints ensures a fair comparison between top-view and back-view performance.

4.3 Training Strategy

The models were trained using labeled datasets consisting of BCS-annotated images. Standard preprocessing and data augmentation techniques were applied to improve generalization, including image resizing, normalization, and geometric transformations such as flipping and rotation.

The EfficientNet-B0 models were initialized using ImageNet-pretrained weights. During training, most pretrained convolutional layers were frozen, while the final classification layer was fine-tuned for Body Condition Score (BCS) classification. This transfer learning strategy reduced training complexity and improved convergence on limited agricultural datasets.

The models were trained using the Adam optimizer with categorical cross-entropy loss. Validation accuracy was monitored during training to reduce overfitting and evaluate model generalization performance.

EfficientNet-B0 was selected due to its lightweight architecture, reduced computational complexity, and strong performance on image classification tasks.

4.4 Prediction Pipeline

The overall prediction pipeline of the system is as follows:

- 1. Collection of real-world images from dairy farms
- 2. Separation of images into top-view and back-view categories
- 3. Mapping of corresponding images for each cow
- 4. Processing using the respective trained models
- 5. Generation of predicted BCS values
- 6. Storage of predictions in CSV format

This pipeline ensures systematic processing of real-world data and enables structured analysis of model outputs.

4.5 Evaluation Framework

The system is evaluated using real-world farm data. For each cow, predictions from both models are compared with veterinarian-provided ground truth.

The evaluation is performed under three scenarios:

- **Full Agreement:** Both predictions match the ground truth.
- **Partial Agreement:** Only one prediction matches.
- **No Agreement:** Neither prediction matches.

This evaluation framework provides insights into model reliability, viewpoint effectiveness, and system performance in practical conditions.

5 Dataset

5.1 Back-View Dataset

The back-view model is trained using a publicly available dataset obtained from the SciDB repository. This dataset contains approximately 53,000 images of dairy cattle collected from real farm environments.

Each image is annotated with a Body Condition Score (BCS) assigned by expert veterinarians. The dataset includes fine-grained BCS categories, typically ranging from 3.25 to 4.25 in increments of 0.25. The large size and diversity of the dataset make it suitable for training deep learning models with strong generalization capability.

The dataset captures variations in lighting, background, and animal posture, which helps the model learn robust feature representations for real-world conditions.

5.2 Top-View Dataset

The top-view dataset is constructed through web scraping from an online academic resource. Initially, approximately 400 labeled images were collected along with their corresponding BCS values.

To address the limitation of small dataset size, extensive data augmentation techniques were applied to increase diversity and improve model performance:

- Horizontal flipping
- Rotation and affine transformations
- Color jitter

After augmentation, the dataset size increased to approximately 8,000 images. This process enables the model to generalize better despite the limited availability of original data. Transfer learning was particularly beneficial for the top-view dataset due to the limited number of original samples.

5.3 Real-World Dataset

To evaluate the practical applicability of the proposed system, a real-world dataset was collected from nearby dairy farms. Images were captured using a standard camera under natural farm conditions.

For each cow, the dataset includes:

- One top-view image
- One back-view image
- A BCS value assigned by a professional veterinarian (ground truth)

This results in a paired dataset where each sample consists of both viewpoints along with an expert label.

The real-world dataset introduces several challenges, including variations in lighting, background clutter, and animal movement. These factors make the evaluation more realistic and help assess the robustness of the proposed system beyond controlled datasets.



(a) Top view (Sample 1)



(b) Back view (Sample 1)



(c) Top view (Sample 2)



(d) Back view (Sample 2)

Fig. 1: Sample images from the real-world dataset showing top-view and back-view perspectives of dairy cattle

5.4 Summary

The use of multiple datasets enables a comprehensive evaluation of the system. The large-scale back-view dataset supports robust model training, while the augmented top-view dataset addresses data scarcity. The real-world dataset provides a practical validation framework, ensuring that the proposed models perform effectively under real farm conditions.

6 Results and Discussion

6.1 Experimental Setup

The trained models were evaluated using real-world data collected from nearby dairy farms. A total of approximately 120 cows were considered in this study. For each cow, multiple images were captured from both viewpoints, resulting in 307 top-view images and 320 back-view images.

The collected images were processed using the trained models developed in the previous stages. Both top-view and back-view images were processed using

Table 1: Model performance on validation datasets

Model	Validation Accuracy
EfficientNet-B0 (Top View)	0.6924
EfficientNet-B0 (Back View)	0.7831

viewpoint-specific EfficientNet-B0 models fine-tuned using transfer learning. The predicted Body Condition Scores (BCS) for each image were stored in structured CSV files for further analysis.

6.2 Prediction Pipeline

The evaluation pipeline consists of the following steps:

1. Collection of real-world images from dairy farms
2. Separation of images into top-view and back-view categories
3. Mapping of corresponding top-view and back-view images for each cow
4. Processing of images using the respective trained models
5. Generation of predicted BCS values for each image
6. Storage of predictions in CSV format for systematic analysis

At this stage, the system has successfully generated predicted BCS values for all collected samples.

6.3 Results

The trained models were applied to the real-world dataset collected from nearby dairy farms. For each cow, both top-view and back-view images were processed using their respective models to generate predicted Body Condition Score (BCS) values.

The back-view model achieved higher validation accuracy compared to the top-view model. This suggests that back-view images contain more discriminative anatomical information such as tail-head prominence, hip structure, and rear fat distribution, which are important indicators for Body Condition Score estimation.

A subset of the predicted results is presented in Table 2. The table includes predictions from both models along with the corresponding BCS annotations provided independently by two veterinarians for each viewpoint. This structured comparison enables direct observation of model outputs alongside expert evaluations.

The results demonstrate that the system is capable of processing real-world data and generating consistent BCS predictions across different viewpoints. Variations between top-view and back-view predictions can be observed in certain cases, reflecting differences in visual information captured by each perspective.

A subset of the results is presented in Table 2.

The top-view model achieved approximately 69 percent validation accuracy, while the back-view model achieved approximately 78 percent validation accuracy. The results indicate that viewpoint selection plays a significant role in automated BCS estimation.

Table 2: Sample predicted BCS values from top-view and back-view models

Cow ID	Top Pred	Back Pred	Vet1 (Top)	Vet1 (Back)	Vet2 (Top)	Vet2 (Back)
1	4.0	3.75	4.0	3.75	4.0	3.75
2	3.75	3.5	3.75	3.5	3.75	3.5
3	3.75	3.75	3.75	3.75	3.75	3.75
4	3.25	4.25	3.5	4.0	3.5	4.0
5	3.75	3.75	3.75	3.75	3.75	3.75

6.4 Ground Truth Validation

To ensure reliable evaluation, the collected dataset was annotated independently by two professional veterinarians. Each expert assigned Body Condition Scores (BCS) separately for both top-view and back-view images of each cow.

The use of multiple annotators reduces subjectivity and improves the robustness of the ground truth labels. The availability of viewpoint-specific annotations enables a more detailed evaluation of model performance by allowing direct comparison between predictions and expert assessments for each view.

These annotations form the basis for evaluating the agreement between model predictions and expert judgments, as well as analyzing consistency across different viewpoints.

6.5 Discussion

The current stage of the work demonstrates the successful deployment of trained deep learning models on real-world farm data. The system is capable of generating BCS predictions from both top-view and back-view images, indicating its practical applicability in agricultural environments.

The availability of predictions for all collected samples confirms the robustness of the data processing pipeline and the generalization capability of the trained models.

Experimental evaluation demonstrates that the back-view model outperforms the top-view model in terms of validation accuracy. The superior performance of the back-view perspective may be attributed to clearer visibility of anatomical structures associated with fat deposition.

The use of EfficientNet-B0 with transfer learning and fine-tuning enabled effective feature extraction despite limited dataset size.

Although the top-view model achieved comparatively lower accuracy, it still demonstrated the capability to estimate BCS under real-world farm conditions.

The results confirm the effectiveness of viewpoint-specific deep learning models for automated livestock monitoring.

7 Conclusion

In this work, a multi-view deep learning framework for automated Body Condition Score (BCS) estimation in dairy cattle has been proposed and implemented. The system utilizes two distinct viewpoints—top view and back view—to capture complementary anatomical features of the animal and generate BCS predictions using dedicated models.

Both the top-view and back-view models are based on the EfficientNet-B0 architecture and utilize transfer learning with fine-tuning of the final classification layer. Each model learns to extract relevant visual features from its respective viewpoint and predict the corresponding Body Condition Score (BCS). This dual-model design enables the system to leverage complementary information from different perspectives of the animal, improving the overall robustness of BCS estimation.

The models were trained on publicly available datasets and subsequently evaluated on real-world data collected from dairy farms. The system successfully generated BCS predictions for all collected samples, demonstrating its practical applicability. Validation using veterinarian-provided ground truth demonstrated the effectiveness of the proposed framework under real-world farm conditions.

Overall, the proposed framework highlights the importance of multi-view analysis and real-world validation in developing robust and scalable solutions for automated livestock monitoring within Agricultural Cyber-Physical Systems.

The back-view model achieved approximately 78 percent validation accuracy, outperforming the top-view model which achieved approximately 69 percent validation accuracy. The findings suggest that back-view images provide more reliable anatomical cues for automated Body Condition Score estimation. The use of a unified EfficientNet-B0 architecture ensured fair comparison across viewpoints while maintaining computational efficiency.

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